

LECTURE 2

GROUP AND RING THEORY

2. FURTHER APPLICATIONS OF THE ORBIT-STABILISER THEOREM

We first give some important applications of the class equation of a finite group G (cf. Theorem 1.25). Before that, observe that $x \in Z(G)$ if and only if the conjugacy class $\text{ccl}(x)$ has just one element, namely x itself. Therefore G is the disjoint union of $Z(G)$ and all conjugacy classes having more than one element. Hence, the class equation can be expressed as

$$|G| = |Z(G)| + \sum_{x \in C} |G : C_G(x)|,$$

where C contains exactly one element from each conjugacy class with more than one element.

The following is an important result in the theory of finite groups of prime power order.

Theorem 2.1. *Let G be a finite group of order p^n , where p is a prime and $n \in \mathbb{N}$. Then*

- (i) G has a non-trivial centre $Z(G)$;
- (ii) $Z(G) \cap K$ is non-trivial for any non-trivial normal subgroup K of G ;
- (iii) if H is a proper subgroup of G , then H is properly contained in $N_G(H)$. Hence if H is a subgroup of order p^{n-1} , then $H \trianglelefteq G$.

Proof. (i) Consider the class equation of G :

$$(2.1) \quad |G| = p^n = |Z| + \sum_{x \in C} |G : C_G(x)|,$$

where $Z = Z(G)$, and C is a subset of G having exactly one element from each conjugacy class not contained in Z . By Lagrange's Theorem, we have $|C_G(x)|$ divides p^n for every $x \in G$, and if $x \notin Z$, then $|C_G(x)| < p^n$. Hence p divides $|G : C_G(x)|$ for every $x \notin Z$. Therefore p divides $\sum_{x \in C} |G : C_G(x)|$. It follows now from the class equation (2.1) that p divides $|Z|$. This proves that Z is not trivial.

(ii) Since $G = Z \cup (\cup_{x \in C} \text{ccl}(x))$ is a disjoint union,

$$K = G \cap K = (Z \cap K) \cup (\cup_{x \in C} \text{ccl}(x) \cap K)$$

is also a disjoint union. Hence

$$(2.2) \quad |K| = |Z \cap K| + \sum_{x \in C} |\text{ccl}(x) \cap K|.$$

If $x \in K$, then $\text{ccl}(x) \subseteq K$ since $K \trianglelefteq G$. Hence, for every $x \in C$, it follows that $\text{ccl}(x) \cap K$ is either empty or equal to $\text{ccl}(x)$, and therefore $|\text{ccl}(x) \cap K|$ is either zero or equal to $|G : C_G(x)|$. Hence p divides $\sum_{x \in C} |\text{ccl}(x) \cap K|$. As p divides $|K|$, it follows from (2.2) that p divides $|Z \cap K|$. This proves that $Z \cap K$ is not trivial.

(iii) Since $H \subseteq N_G(H)$, to show that $H \subsetneq N_G(H)$, we just need to show that $H \neq N_G(H)$. To this end, let M be a maximal normal subgroup of G contained in H (i.e. if $M \leq L \trianglelefteq G$ with $L \subseteq H$ then $L = M$). We may assume that $H \neq 1$, as otherwise the result is clear since

$N_G(1) = G$. Then the quotient group G/M is of order p^r for some¹ $1 \leq r \leq n$. Hence by (i) the group G/M has a non-trivial centre, say L/M . Since $L/M \trianglelefteq G/M$, it follows by the Correspondence Theorem that $L \trianglelefteq G$. Now L is not contained in H , since M is a maximal normal subgroup of G contained in H , and $M \subsetneq L$.

Thus, to finish the proof of the first part of the statement, it suffices to show that $L \subseteq N_G(H)$. Let $h \in H$, $\ell \in L$. Since $L/M = Z(G/M)$, we have $(hM)(\ell M) = (\ell M)(hM)$. Thus $\ell^{-1}h\ell \in hM \subseteq H$. Therefore $L \subseteq N_G(H)$.

Finally, if H is a subgroup of order p^{n-1} , then $N_G(H)$ must be of order p^n . Hence $N_G(H) = G$ and so $H \trianglelefteq G$. $\square$

The following (which was first discovered by Cauchy and Frobenius) has numerous applications, also in combinatorics. For a fixed $g \in G$, let $X_g := \{x \in X \mid gx = x\}$, where X is a G -set.

Theorem 2.2 (Burnside's Lemma). *Let G be a finite group acting on a finite set X . Then the number k of orbits in X under G is*

$$k = \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

Proof. Let $S := \{(g, x) \in G \times X \mid gx = x\}$. For any fixed $g \in G$ the number of ordered pairs (g, x) in S is exactly $|X_g|$. Similarly, for any fixed $x \in X$, we have that $|G_x|$ is the number of ordered pairs (g, x) in S . Hence,

$$\sum_{g \in G} |X_g| = |S| = \sum_{x \in X} |G_x|.$$

By the Orbit-Stabiliser Theorem (Theorem 1.24), we have $|Gx| = |G : G_x| = |G|/|G_x|$. Hence

$$\sum_{x \in X} |G_x| = |G| \sum_{x \in X} \frac{1}{|Gx|} = |G| \sum_{a \in C} \sum_{x \in Ga} \frac{1}{|Gx|},$$

where C is a subset of X containing exactly one element from each orbit. Now $|Gx| = |Ga|$ for every $x \in Ga$. Thus,

$$\sum_{x \in Ga} \frac{1}{|Gx|} = 1.$$

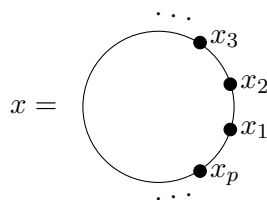
Therefore $\sum_{x \in X} |G_x| = |G| \cdot k$, which yields the result. $\square$

The following example gives a nice illustration of Burnside's Lemma.

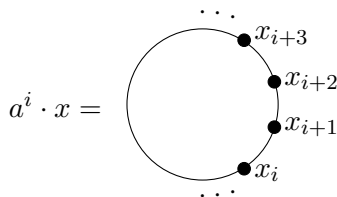
Example 2.3. Find the number of different necklaces with p beads, for p a prime, where the beads can have any of n different colours. [Assume that you cannot flip the necklaces, for instance if the beads are different on the reverse side.]

¹We actually know that $r < n$, since $M \neq 1$ by (i); indeed, we have $1 \neq Z(H) \subseteq M$. But the fact that $r < n$ is not necessary for the rest of the proof.

Solution: Let X be the set of all possible necklaces arrangements/colourings. Clearly $|X| = n^p$. Let



be a necklace. Let $G = \langle a \rangle$ be the cyclic group of order p . Then X is a G -set under the action



where subscripts are modulo p .

Clearly the action of any fixed element $a^i \in G$ on a given necklace yields the same necklace. (The beads are just being permuted cyclically.) So the number of orbits is the same as the number of different necklaces. We now compute X_g for each $g \in G$. First

$$X_e = \{x \in X \mid e \cdot x = x\} = X.$$

So $|X_e| = n^p$. Let $g \neq e$. Since G is cyclic of prime order, the element g generates G . Then

$$\begin{aligned} X_g &= \{x \in X \mid g \cdot x = x\} \\ &= \{x \in X \mid g^i \cdot x = x \quad \forall i \in \{0, 1, \dots, p-1\}\} \\ &= \{x \in X \mid a \cdot x = x \quad \forall a \in G\}. \end{aligned}$$

Therefore X_g consists of all those necklaces which are unchanged by any permutation and these are precisely those which are of one colour. Hence $|X_g| = n$ for all $g \in G \setminus \{e\}$. Then Burnside's Lemma then gives the number of different necklaces as

$$\frac{1}{p} \left[n^p + n + \overset{p-1}{\dots} + n \right] = \frac{1}{p} [n^p + n(p-1)].$$

This above sort of counting problem is part of what is known as Pólya counting.

3. SYLOW THEOREMS AND APPLICATIONS

3.1. Sylow theorems. Recall from Algebraic Structures the decomposition of a finite abelian group A , which provides a complete description of the structure of A . There is no such general result for finite non-abelian groups. However, the Sylow theorems, which occupy an important place in the theory of finite groups, are a powerful tool in studying the structure of finite non-abelian groups. In particular, the existence or non-existence of a simple group of a given order can sometimes be determined by application of the Sylow theorems, among other machinery.

Recall the following from Algebraic Structures.

Definition 3.1. Let p be a prime. A group G is called a p -group if the order of every element in G is some power of p . Likewise, a subgroup H of any group G is called a p -subgroup of G if the order of every element in H is some power of p .

Example 3.2. For $n \in \mathbb{N}$ and p a prime, let C_{p^n} denote the cyclic group of order p^n . [If you want, you can think of C_{p^n} as $\mathbb{Z}/p^n\mathbb{Z}$.] The infinite group $C_p \oplus C_{p^2} \oplus C_{p^3} \oplus \cdots$ is a p -group, as is $C_p \times C_p \times C_p \times \cdots$; note that the direct sum means that the elements in the group consist of sequences with only finitely many non-zero entries. The finite group $C_p \times C_p$ is a p -group.

We now consider the following new concept.

Definition 3.3. Let G be a finite group, and let p be a prime. Let $m \in \mathbb{N} \cup \{0\}$ be such that $p^m \mid |G|$ but $p^{m+1} \nmid |G|$. Then any subgroup of G of order p^m is called a *Sylow p -subgroup* of G .

Example 3.4. Let p be a prime. If $G = C_p \times C_p$, then G itself is the only Sylow p -subgroup. If $G = C_{p^2} \times C_q$, for a prime $q \neq p$, then C_{p^2} is the only Sylow p -subgroup of G , and C_q is the only Sylow q -subgroup of G . If $G = \text{GL}_2(\mathbb{F}_p)$, one can check that $|G| = (p^2 - p)(p^2 - 1) = p(p + 1)(p - 1)^2$. Then a Sylow p -subgroup of G is the subgroup of diagonal scalar matrices $\left\{ \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \mid \lambda \in \mathbb{Z}_p \right\} \cong C_p$. Note that if q is a prime dividing $(p + 1)(p - 1)^2$, then G will have a Sylow q -subgroup.